

1 Problem 1

First, normalize the density function to find the pdf ($p(x)$):

$$\int_0^1 f(x) = \int_0^1 (1+x)dx = \left[x + \frac{x^2}{2} \right]_0^1 = 1 + \frac{1}{2} = \frac{3}{2} \quad (1)$$

$$p(x) = \frac{2}{3}f(x) = \frac{2}{3}(1+x) \quad (2)$$

Next, use Eq. 1 from the probability and stats lecture for a pdf, $p(x)$, in the range of (a, b):

$$\mathbb{E}[g(X)] = \int_a^b g(x)p(x)dx \quad (3)$$

$$\mathbb{E}[g(X)] = \int_0^1 g(x)p(x)dx = \int_0^1 (x+cx^2)\frac{2}{3}(1+x)dx = \frac{2}{3} \int_0^1 cx^3 + (1+c)x^2 + x \, dx \quad (4)$$

$$\int_0^1 cx^3 \, dx = \left[\frac{c}{4}x^4 \right]_0^1 = \frac{c}{4} \quad (5)$$

$$\int_0^1 (1+c)x^2 = \left[(1+c)\frac{x^3}{3} \right]_0^1 = \frac{1+c}{3} \quad (6)$$

$$\int_0^1 x \, dx = \left[\frac{x^2}{2} \right]_0^1 = \frac{1}{2} \quad (7)$$

Which yields an expectation for $g(x)$ of:

$$\mathbb{E}[g(X)] = \frac{2}{3} \left(\frac{c}{4} + \frac{1+c}{3} + \frac{1}{2} \right) = \frac{7c}{18} + \frac{5}{9} \quad (8)$$

Next, we will calculate the expectation value of $(g(x))^2$:

$$\mathbb{E}[(g(X))^2] = \int_0^1 (g(x))^2 f(x) \, dx = \int_0^1 (x+cx)^2 \frac{2}{3} (1+x) = \quad (9)$$

$$= \frac{2}{3} \int_0^1 c^2 x^5 + (c^2 + 2c)x^4 + (2c+1)x^3 + x^2 \, dx \quad (10)$$

$$\int_0^1 c^2 x^5 \, dx = \left[c^2 \frac{x^6}{6} \right]_0^1 = \frac{c^2}{6} \quad (11)$$

$$\int_0^1 (c^2 + 2c)x^4 \, dx = \left[(c^2 + 2c)\frac{x^5}{5} \right]_0^1 = \frac{c^2 + 2c}{5} \quad (12)$$

$$\int_0^1 (2c+1)x^3 \, dx = \left[(2c+1)\frac{x^4}{4} \right]_0^1 = \frac{2c+1}{4} \quad (13)$$

$$\int_0^1 x^2 dx = \left[\frac{x^3}{3} \right]_0^1 = \frac{1}{3} \quad (14)$$

Which yields an expectation for $(g(x))^2$ of:

$$\mathbb{E}[(g(X))^2] = \frac{2}{3} \left(\frac{c^2}{6} + \frac{c^2 + 2c}{5} + \frac{2c + 1}{4} + \frac{1}{3} \right) = \frac{11c^2}{45} + \frac{3c}{5} + \frac{7}{18} \quad (15)$$

We then use Eq. 11 from the same lecture to calculate the variance:

$$\sigma_x^2 = \mathbb{E}[X^2] - (\mathbb{E}[X])^2 \quad (16)$$

$$\sigma_g^2 = \left(\frac{11c^2}{45} + \frac{3c}{5} + \frac{7}{18} \right) - \left(\frac{7c}{18} + \frac{5}{9} \right)^2 \quad (17)$$

$$= \left(\frac{11c^2}{45} + \frac{3c}{5} + \frac{7}{18} \right) - \left(\frac{49c^2}{324} + \frac{35c}{81} + \frac{25}{81} \right) \quad (18)$$

$$= \left(\frac{11}{45} - \frac{49}{324} \right) c^2 + \left(\frac{3}{5} - \frac{35}{81} \right) c + \left(\frac{7}{18} - \frac{25}{81} \right) \quad (19)$$

$$= \frac{396 - 245}{1620} c^2 + \frac{243 - 173}{405} c + \frac{63 - 50}{162} \quad (20)$$

$$= \frac{151}{1620} c^2 + \frac{68}{405} c + \frac{13}{162} \quad (21)$$

We are trying to minimize the variance with respect to c , so we set the derivative equal to zero:

$$\frac{d\sigma_g^2}{dc} = 0 \quad (22)$$

$$\frac{d}{dc} \left(\frac{151}{1620} c^2 + \frac{68}{405} c + \frac{13}{162} \right) = 0 \quad (23)$$

$$\frac{151c}{810} + \frac{68}{405} = 0 \quad (24)$$

$$c = -\frac{68}{405} \frac{810}{151} = -\frac{136}{151} \quad (25)$$

To minimize the variance of $g(x)$, we should set c to

$$c = -\frac{136}{151} \quad (26)$$

2 Problem 2

2.1 Results

The program is in the attached jupyter notebook. Here are my results. I told AI to format my python results into a nice $\text{\LaTeX}$ table.

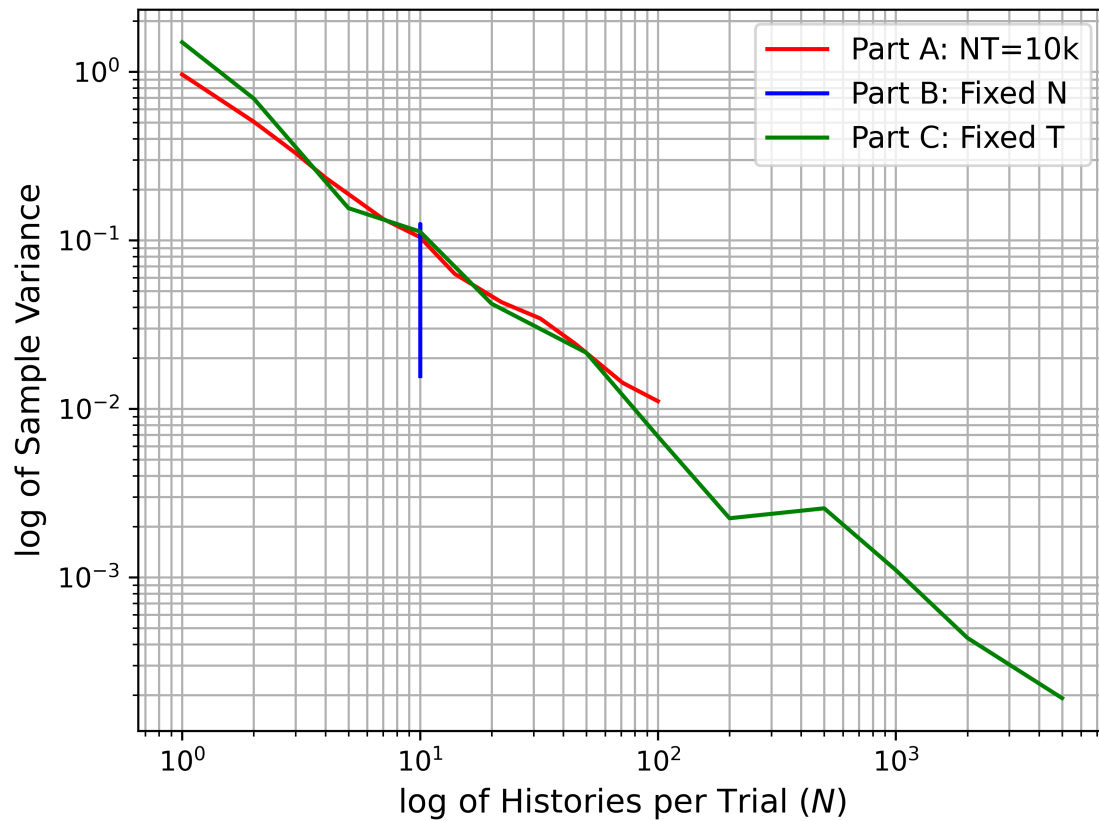


Figure 1: Variance with respect to number of histories per trial

Table 1: Results for Part A

Histories per trial (N)	Number of Trials (T)	Mean	Variance
100	100	1.001 46	$1.109\,67 \times 10^{-2}$
71	141	0.995 46	$1.428\,15 \times 10^{-2}$
45	222	1.014 77	$2.428\,38 \times 10^{-2}$
32	312	1.006 37	$3.437\,55 \times 10^{-2}$
22	455	1.001 59	$4.295\,68 \times 10^{-2}$
14	714	0.994 11	$6.319\,23 \times 10^{-2}$
10	1000	1.002 70	$1.043\,15 \times 10^{-1}$
7	1429	1.009 50	$1.339\,34 \times 10^{-1}$
4	2500	0.973 46	$2.355\,30 \times 10^{-1}$
3	3333	0.987 08	$3.292\,99 \times 10^{-1}$
2	5000	1.006 44	$5.055\,61 \times 10^{-1}$
1	10000	1.000 39	$9.647\,07 \times 10^{-1}$

Table 2: Results for Part B

Histories per trial (N)	Number of Trials (T)	Mean	Variance
10	10	1.005 71	$1.566\,08 \times 10^{-2}$
10	20	0.891 99	$7.774\,78 \times 10^{-2}$
10	30	1.045 62	$7.488\,02 \times 10^{-2}$
10	40	1.034 13	$7.708\,72 \times 10^{-2}$
10	50	1.024 38	$1.250\,86 \times 10^{-1}$
10	100	1.016 08	$1.046\,76 \times 10^{-1}$
10	200	0.967 87	$1.032\,52 \times 10^{-1}$
10	500	0.999 86	$1.097\,56 \times 10^{-1}$
10	1000	1.020 49	$1.096\,11 \times 10^{-1}$

Table 3: Results for Part C

Histories per trial (N)	Number of Trials (T)	Mean	Variance
1	20	1.108 34	1.497 79
2	20	0.940 39	$6.939\,88 \times 10^{-1}$
5	20	1.001 56	$1.552\,74 \times 10^{-1}$
10	20	0.969 30	$1.126\,75 \times 10^{-1}$
20	20	1.030 84	$4.203\,17 \times 10^{-2}$
50	20	1.006 90	$2.151\,97 \times 10^{-2}$
100	20	0.947 46	$6.853\,08 \times 10^{-3}$
200	20	0.993 89	$2.241\,46 \times 10^{-3}$
500	20	0.984 35	$2.567\,44 \times 10^{-3}$
1000	20	1.002 64	$1.103\,39 \times 10^{-3}$
2000	20	0.994 26	$4.368\,97 \times 10^{-4}$
5000	20	1.001 61	$1.918\,55 \times 10^{-4}$

2.2 Discussion

The central limit theorem states: when independent random variables are added, their properly normalized sum tends towards a normal distribution even if the original random variables are not normally distributed. Additionally, the sample mean, $\langle \bar{x} \rangle$, is normally distributed about the true mean, m_x , with a variance of S_x^2/T .

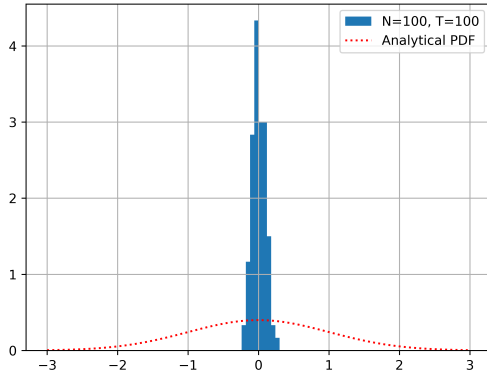
From the plot, we see the results from part A and part C follow the same trend with a slope of roughly negative one in log-log space. The number of trials does not matter when calculating the variance. For part A, we try to simulate the same number of total trials by proportionally decreasing the number of trials as we increase the histories per trial. For part C, we fix the number of trials and increase the histories per trial. Since we observe the same trend in sample variance independent of the number of trials, we conclude the variance only scales with respect to the number of histories per trial.

Regarding the central limit theorem, each trial acts like an independent random event. For the trials with histories per trial, we see the variance decreases. For part A with $N=1$ and $T=10000$, each trial acts as a single sample. As such, the mean for each trial is equal to the single event from that trial. So, when we calculate the total mean, the total mean will be the same as if we only ran one trial with 10000 histories per trial. The trials here are easily compared to a single random event. When we look at trials with more histories per trial, we anticipate trial mean gets closer to the true mean because each trial is generate with more random numbers, which have the same mean.

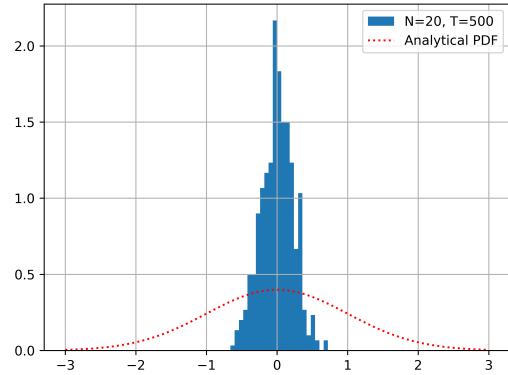
3 Question 3

3.1 Histogram Comparison

I want to start by saying the question is poorly worded. The question prompt is; "Compare the normalized histogram of these T samples with the corresponding PDF of the Gaussian distribution." Since "corresponding" is an adjective on "PDF", the statement means the for the Gaussian distribution, plot the corresponding PDF. Which Gaussian distribution? The only information we have about the Gaussian is "the". I would assume this means the standard, normal Gaussian in the same way that somebody taking "the logarithm" of a number is referring to the natural logarithm. I think the question was asking the compare the samples with the "PDF of the corresponding (non-standard) Gaussian distribution," but this is not what the statement means.



(a) Histogram: Experiment 1



(b) Histogram: Experiment 2

We see both experiments look like the standard normal distribution, but symmetrically "squeezed" together from both sides. This yields a normal distribution with a higher mean, but lower variance for both trials. If I plotted the corresponding normal distribution, we would see these samples still follow the corresponding normal distribution, but not "the" normal distribution. For experiment 1, we see the non-standard normal distribution has a higher mean than the standard normal distribution, but a proportionally smaller variance/standard deviation. For experiment 2, we also see a higher mean and smaller variance/standard deviation than the standard normal distribution. Since the number of histories per trial is greater in experiment 1, we see a higher mean because each trial uses 100 independent samples to generate a single macro-sample (trial). This is what the central limit theorem states. We see the same consequence in experiment 2, but to a lesser extent because there are only 20 samples used to generate a macro-sample (trial). If we did another experiment with $N=1$, we would see the standard normal distribution.

3.2 Skewness Test

I told AI to format my results into a nice looking table.

Table 4: Statistical Values and Significance Thresholds (T=100) for Experiment 1

Calculated Values		Significance Thresholds (T=100)		
Method	Value	Level	$C^2 \geq$	$ C \geq$
From Lecture Slides	0.016971	Doubtful	0.21496	0.46364
Scipy Biased (default)	0.017229	Meaningless	0.44184	0.66471
Scipy Unbiased	0.017492			

Table 5: Statistical Values and Significance Thresholds (T=500) for Experiment 2

Calculated Values		Significance Thresholds (T=500)		
Method	Value	Level	$C^2 \geq$	$ C \geq$
From Lecture Slides	-0.024156	Doubtful	0.045459	0.21321
Scipy Biased (default)	-0.024229	Meaningless	0.093437	0.30567
Scipy Unbiased	-0.024302			

From the skewness test, I obtained the skewness values as given in the tables above. For both experiments, all calculated skewness coefficients are an order of a less than the threshold values for doubtful and meaningless. Therefore, we fail to reject the null hypothesis and can state our distribution is statistically indistinguishable from a known, non-skewed distribution. These results show the data are very symmetrically distributed about the mean. Scipy uses the biased sample variance by default, but you can set the function to calculate the unbiased skew coefficient.

3.3 Shapiro-Wilk Test

I asked AI to make this table too.

Table 6: Shapiro-Wilk Normality Test Results

Dataset	Test Statistic (W)	p-value
Experiment 1	0.99432	0.95354
Experiment 2	0.99815	0.87174

The critical value for comparison to the null hypothesis is $\alpha = 0.05$. We have p values of 0.95354 and 0.87174, which are an order of magnitude greater than the critical value. The p value is the chance a random distribution would generate our distribution. We have incredibly high p-values: for experiment 1, there is a 95.4% chance a random distribution would generate a distribution like ours; for experiment 2, there is an 87.2% chance a random distribution would generate a distribution like ours. Therefore, we fail to reject the null hypothesis and state our distribution is statistically indistinguishable from a known, random distribution.

4 Problem 4

I, for the last time this homework, asked AI to format my python results into a nice latex table.

Table 7: Comparison of Hypothesis Testing and Confidence Intervals

Metric	Part 1 (<i>t</i>-Distribution)	Part 2 (<i>z</i>-Distribution)
Calculated Test Statistic	2.7386	2.8284
Critical Value (90%)	1.6990	1.6450
Critical Value (95%)	2.0450	1.9600
90% Margin of Error	0.12408	0.11632
90% Confidence Interval	[4.67592, 4.92408]	[4.68368, 4.91632]
95% Margin of Error	0.14935	0.13859
95% Confidence Interval	[4.65065, 4.94935]	[4.66141, 4.93859]

I originally misread the question and calculated the test statistic for the distributions (first row in the table below the headings), but still included this results. We reject the null hypothesis that the factory is producing screws with the target length (null mean) of 5cm according to both sampling events. We are 95% confident in rejecting the null hypothesis as the null mean falls outside both 95% confidence intervals. There might be a screw loose somewhere in the factory.